

HackStreet 2K26 Problem Statements

Problem Statement 1: AI-Powered Code Editor Interface (Frontend Only)

Design and develop a highly interactive frontend interface for an AI-powered code editor that simulates a modern developer environment. The application should include a multi-tab code editor with syntax highlighting for multiple programming languages, an expandable file explorer with nested directory structures, and a resizable layout system similar to professional IDEs.

In addition, integrate a mock AI chat assistant panel capable of displaying contextual suggestions, debugging hints, and code completions based on user input (use static/mock responses). Include an output console that displays simulated execution results, logs, and error messages.

The UI should support advanced interactions such as drag-and-drop file management, theme switching (dark/light), keyboard shortcuts visualization, and responsive layout behavior. The goal is to replicate the feel of tools like VS Code purely on the frontend, focusing on performance, usability, and seamless UI/UX transitions without any real backend or execution engine.

Problem Statement 2: Cryptocurrency Analytics Dashboard (Frontend Only)

Build a comprehensive cryptocurrency dashboard interface that visually represents market data using only static or mock datasets. The application should display real-time-like price updates (simulated), interactive charts (line, candlestick, bar), and a portfolio tracking section showing asset allocation, profit/loss, and historical performance.

Include filtering options for different cryptocurrencies, time ranges, and market categories. The UI should feature advanced data visualization techniques such as animated charts, hover tooltips, and dynamic indicators for trends (bullish/bearish signals).

Additionally, design sections for watchlists, recent transactions, and news highlights (mock content). Emphasize a clean, fintech-style UI with smooth animations, skeleton loaders, and responsive design that adapts across devices. The challenge lies in creating a data-heavy interface that remains intuitive and visually engaging without relying on real APIs.

Problem Statement 3: Hackathon Event Landing Page (Modern UI/UX)

Create a visually stunning and modern landing page for a hackathon event inspired by high-quality tech event websites. The page should include sections such as a hero banner with strong visual impact, event overview, timeline/schedule, prize pool breakdown, sponsor showcase, FAQs, and a registration form (UI only).

Focus on storytelling through design—use animations, gradients, glassmorphism/neumorphism, and micro-interactions to guide user attention. The timeline should be interactive, allowing users to explore different phases of the event. The sponsor section should highlight branding effectively, and the FAQ section should include collapsible/interactive elements.

The registration form should include validation UI, multi-step flow, and confirmation states (mocked). The entire page must be fully responsive, optimized for performance, and visually consistent with modern design trends. The challenge is to balance aesthetics with usability while delivering a production-level frontend experience.

For reference use this:

<https://hackstreet2k26liveone.vercel.app/#about>

Problem Statement 4: Collaborative Whiteboard Application (Frontend Simulation)

Design a frontend-only collaborative whiteboard application that enables users to draw, organize ideas, and simulate team collaboration. The application should include tools such as freehand drawing, shapes, text input, sticky notes, and color selection.

Users should be able to move, resize, and layer elements on the canvas with smooth interactions. Include UI components that simulate collaboration, such as user cursors, participant avatars, and activity indicators (all mock-based).

Enhance the experience with features like zooming, panning, undo/redo actions, and exporting the board as an image or PDF. The interface should feel fluid and intuitive, handling complex interactions without lag.

The goal is to replicate tools like digital whiteboards using only frontend technologies, focusing on state management, rendering performance, and user experience design.